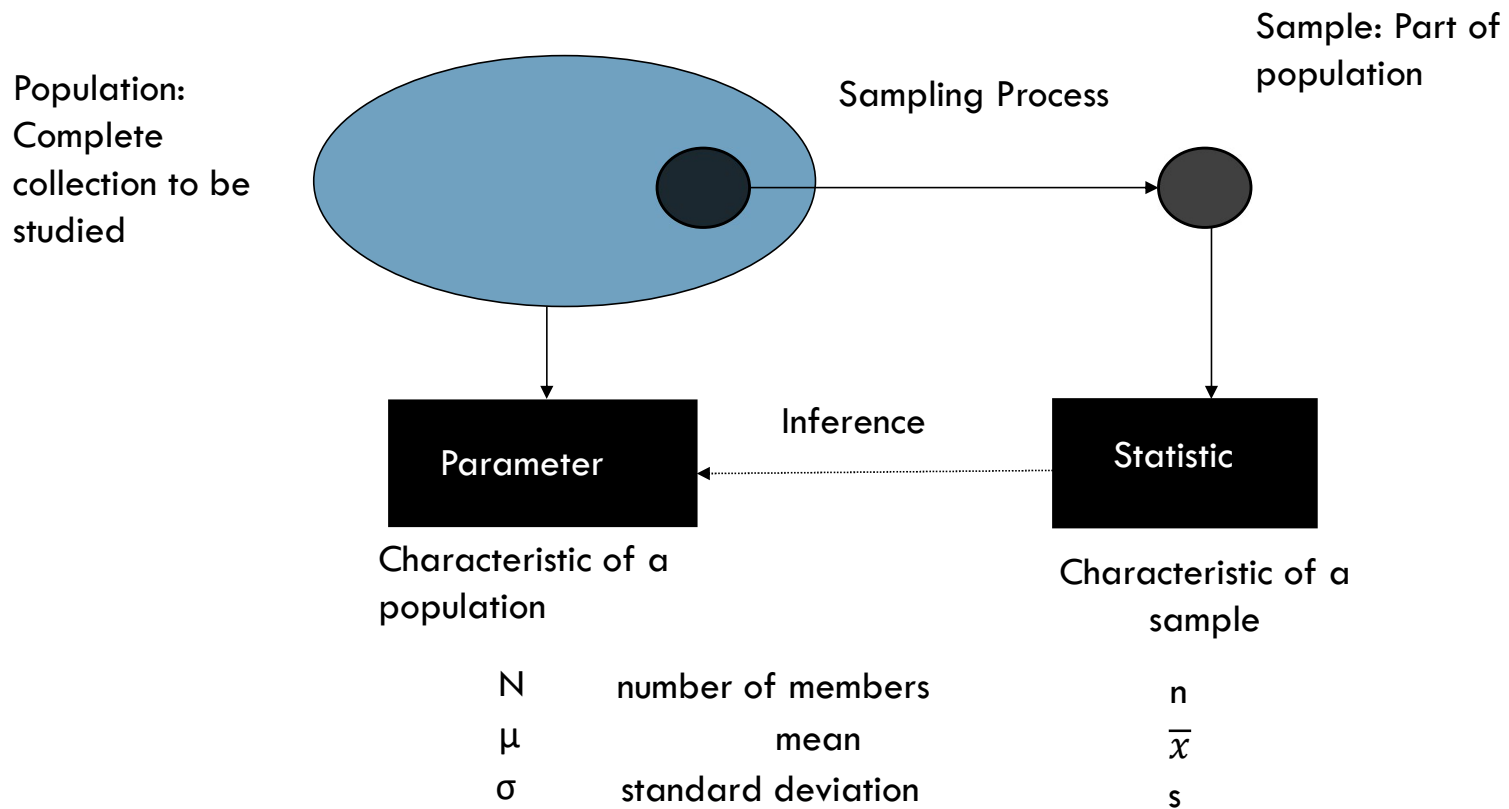


BASIC STATISTICS TERMS

- ❖ Population: The entire group of items or individuals from which data can be collected.
- ❖ Includes metrics like:
 - ❖ N : Total population size.
 - ❖ μ : Population mean (average of all data points in the population).
 - ❖ σ : Population standard deviation (variation within the population).
- ❖ Sample: A subset of the population, selected for analysis.
- ❖ Includes metrics like:
 - ❖ n : Sample size (number of data points in the sample).
 - ❖ $\bar{x}$: Sample mean (average of sample data points).
 - ❖ s : Sample standard deviation (variation within the sample).

BASIC STATISTICAL TERMS



NOTATIONS

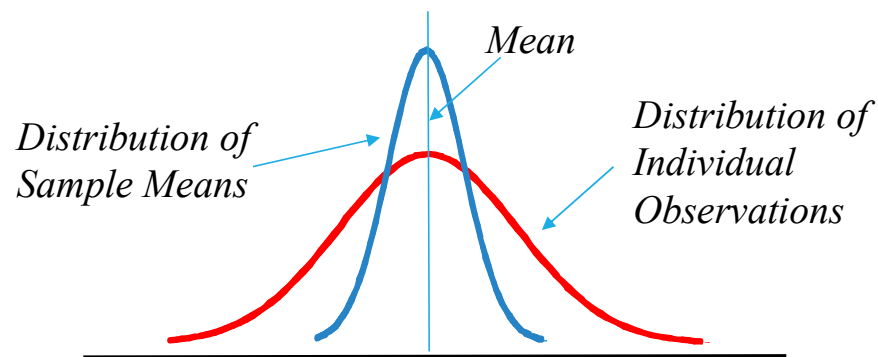
	Population Parameters	Sample Statistics
Mean	μ	$\bar{x}$
Standard Deviation	σ	s
Variance	σ^2	s^2
Proportion of population having an attribute	P	p
Proportion of population not having an attribute	Q ($=1-P$)	q ($=1-p$)
Correlation coefficient	ρ	r
Number of elements	N	n

CENTRAL LIMIT THEOREM (DICE ROLL)

- ❖ When we roll a fair six-sided die, each face (1, 2, 3, 4, 5, 6) is equally likely. This creates a uniform distribution.
- ❖ What Happens with Multiple Dice?:
 - ❖ If we roll one die, the outcome follows a uniform distribution.
 - ❖ If we roll two dice and add their outcomes, the sum (ranging from 2 to 12) starts forming a triangular distribution.
 - ❖ As we keep rolling more dice (e.g., 5, 10 dice), the sum of their values starts to follow a normal distribution.
- ❖ Key Insight:
 - ❖ The more dice we roll and sum, the more the distribution of these sums resembles a normal distribution, even though each die roll is uniformly distributed.

CENTRAL LIMIT THEOREM

- ❖ The Central Limit Theorem states that the sampling distribution of the sample mean (or sum) will approach a normal distribution as the sample size becomes large, regardless of the original population's distribution.
- ❖ Mean and Standard Deviation:
 - ❖ The mean of the sampling distribution equals the population mean μ .
 - ❖ The standard deviation of the sampling distribution, known as the standard error



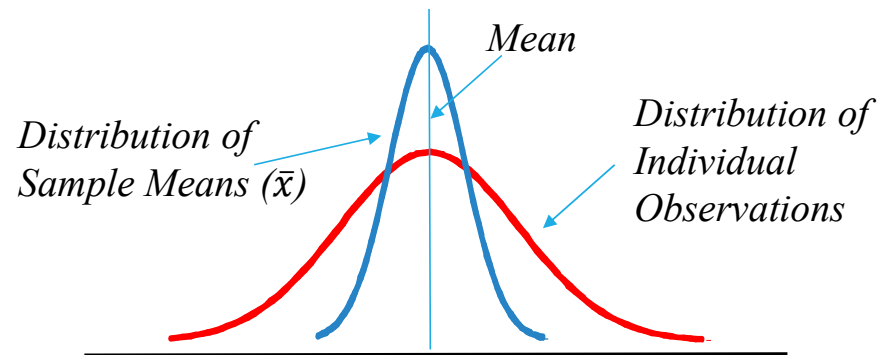
CENTRAL LIMIT THEOREM

- ❖ If a variable has a mean of μ and the variance σ^2 , as the sample size n increases, the sample mean approaches a normal distribution with mean $\mu_{\bar{x}}$ and variance $\sigma_{\bar{x}}^2$.

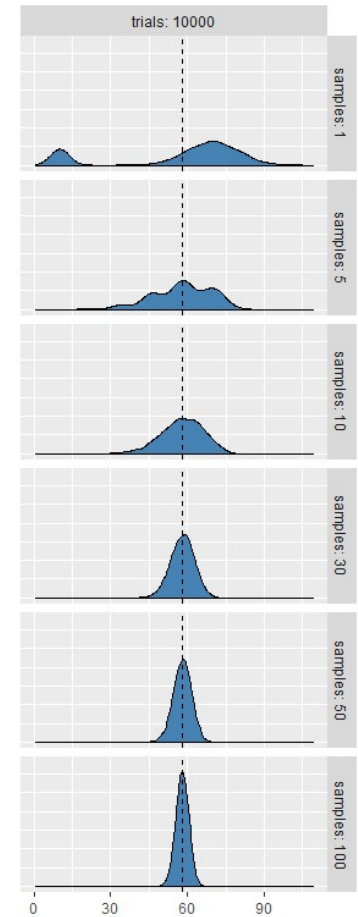
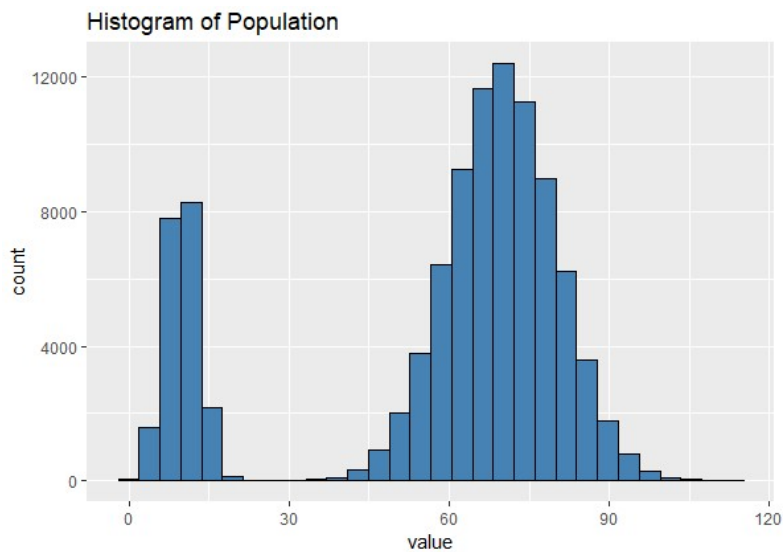
$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{n}$$

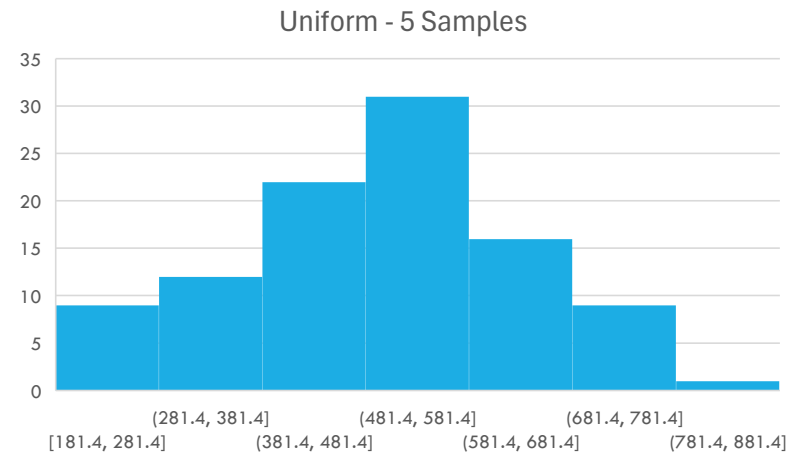
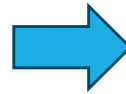
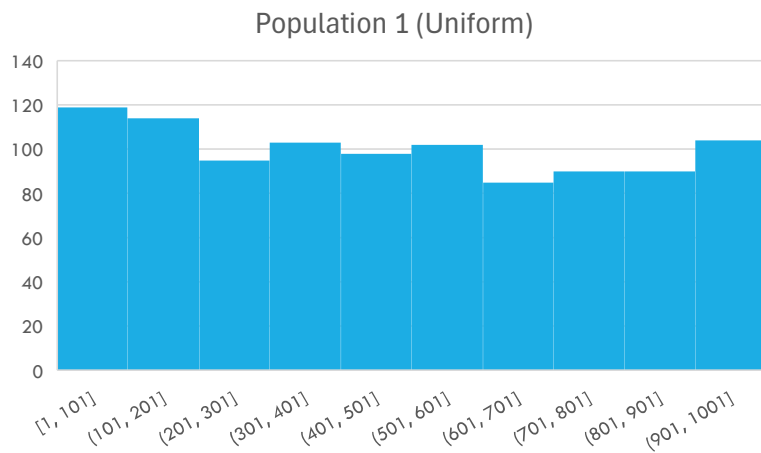
$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$



CENTRAL LIMIT THEOREM (DICE ROLL)



CENTRAL LIMIT THEOREM (EXCEL DEMO)



PROBABILITY

- ❖ Probability is the measure of how likely an event is to occur.
- ❖ It is expressed as a number between 0 and 1 (or 0% to 100%), where:
 - ❖ 0 means the event will not happen.
 - ❖ 1 means the event will certainly happen.
- ❖ Common uses of probability include predicting outcomes in games of chance and real-world

TYPES OF PROBABILITY

❖ Classical (Theoretical) Probability:

- ❖ Based on reasoning and logical calculation.
- ❖ Example: The probability of rolling a 5 on a die is $1/6$ because only one 5 out of six possible outcomes exist.

❖ Experimental (Empirical) Probability:

- ❖ Based on actual experiments and data.
- ❖ Example: Rolling a die 100 times and observing how many times a 5 appears.

❖ Subjective Probability:

- ❖ Based on intuition or personal judgment.
- ❖ Example: The probability of rain tomorrow based on personal experience or weather forecasts.

BASIC DEFINITIONS - PROBABILITY

- ❖ Experiment/Trial: Something done with an expectation of result.
 - ❖ Example: Flipping a coin is a trial.
- ❖ Event: An event is a specific outcome or a group of outcomes from a probability experiment.
 - ❖ Example: Getting heads when flipping a coin.
- ❖ Sample Space: A sample space of an experiment is the set of all possible results of that random experiment.
 - ❖ Example: For a coin flip, the sample space SS is $\{H,T\}$, where:
 - ❖ H = Heads
 - ❖ T = Tails

BASIC DEFINITIONS - PROBABILITY

- ❖ Simple Event: A single outcome.
 - ❖ Example: Flipping a coin and getting Heads (H).
 - ❖ Sample space $S=\{H,T\}$
- ❖ Complex Event: A combination of outcomes, often from multiple trials.
 - ❖ Example: Flipping a coin twice.
 - ❖ Possible outcomes: $\{HH, HT, TH, TT\}$
 - ❖ Here, HH means getting heads twice, HT means getting heads on the first trial and tails on the second, etc.

BASIC DEFINITIONS - PROBABILITY

❖ Simple Event: Rolling a Die:

- ❖ The sample space for rolling a 6-sided die is $S=\{1,2,3,4,5,6\}$.

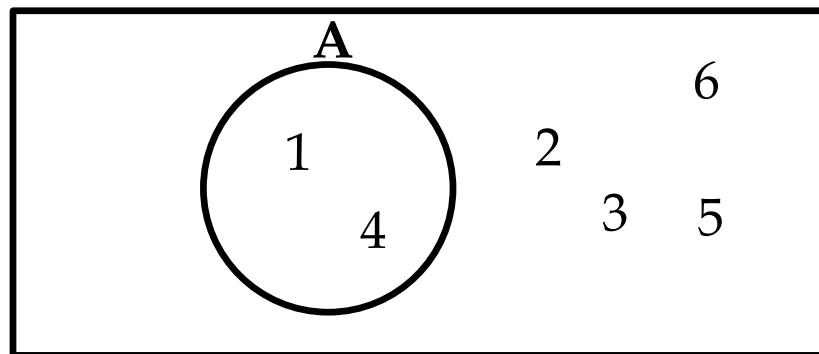
❖ Complex Event:

- ❖ The sample space for rolling two 6-sided dice is:

$\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),$
 $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (5,6),$
 $(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

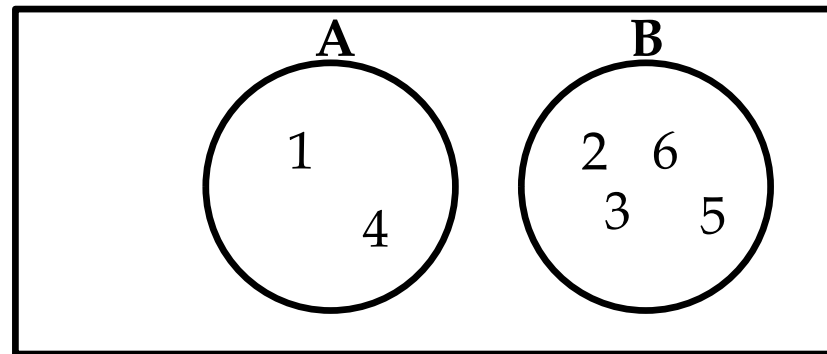
VISUALIZING SAMPLE SPACE WITH VENN DIAGRAMS

- ❖ Venn Diagrams are often used to represent events and their relationships to each other.
- ❖ Example 1: Simple Event A - Probability of rolling a 1 or 4.
- ❖ The sample space for rolling a die is: $(S) = \{1, 2, 3, 4, 5, 6\}$
- ❖ In this case, Event A can be represented in a Venn diagram like this:



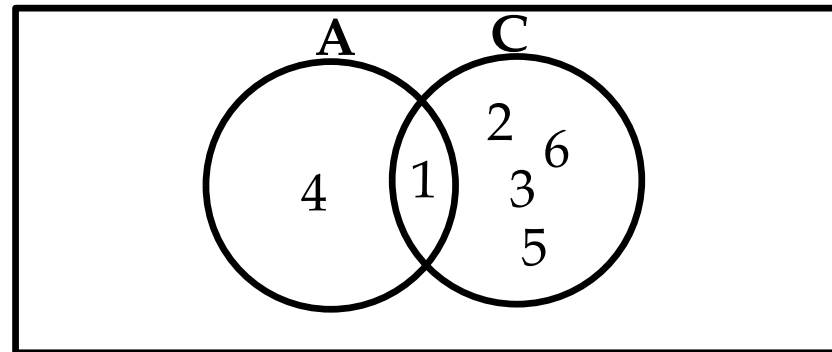
VISUALIZING SAMPLE SPACE WITH VENN DIAGRAMS

- ❖ Non-Overlapping Events A and B
- ❖ Event A: Probability of rolling a 1 or 4.
- ❖ Event B: Probability of rolling a 2, 3, 5, or 6.
- ❖ Event A = {1, 4}
- ❖ Event B = {2, 3, 5, 6}



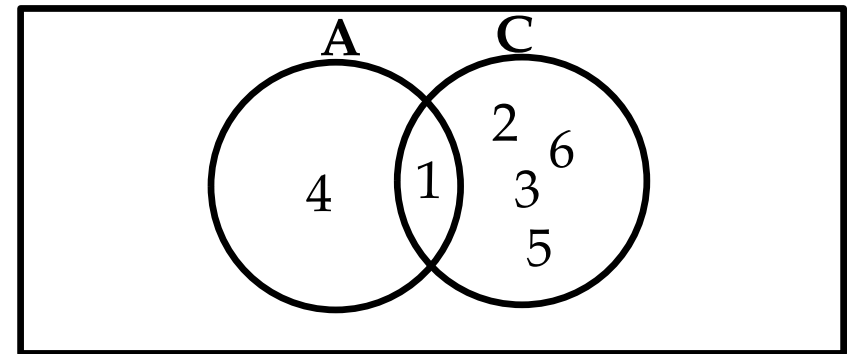
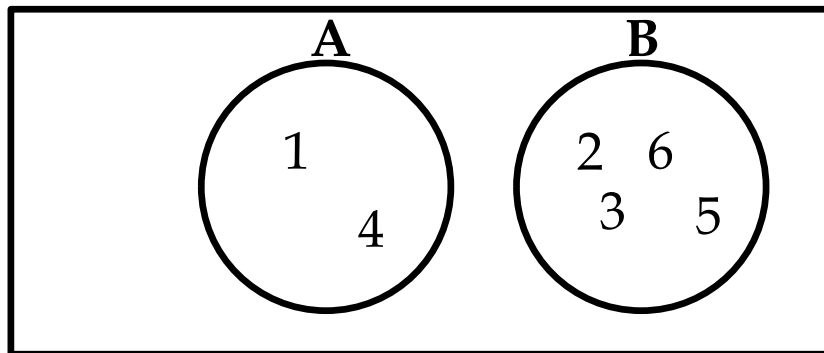
VISUALIZING SAMPLE SPACE WITH VENN DIAGRAMS

- ❖ Example 3: Overlapping Events A and C
- ❖ Event A: Probability of rolling a 1 or 4.
- ❖ Event C: Probability of rolling a 1, 2, 3, 5, or 6.
- ❖ Event A = {1, 4}
- ❖ Event C = {1, 2, 3, 5, 6}



MUTUALLY EXCLUSIVE EVENTS

- ❖ Mutually Exclusive Events: If two events cannot occur at the same time (like Events A and B), they are mutually exclusive.
- ❖ Non-Mutually Exclusive Events: When events can overlap (like Events A and C), they are not mutually exclusive, meaning the outcome can belong to both events.



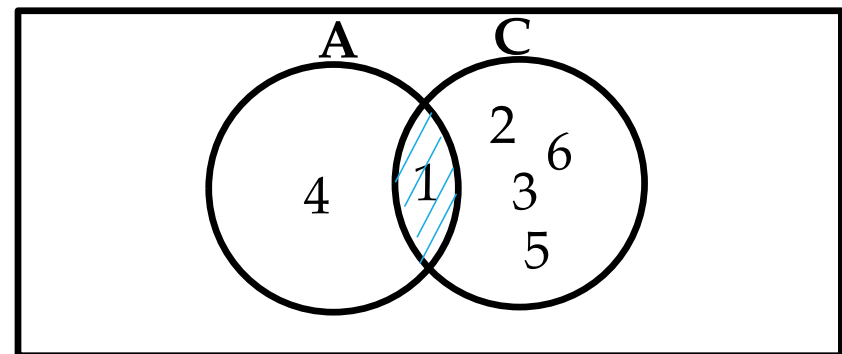
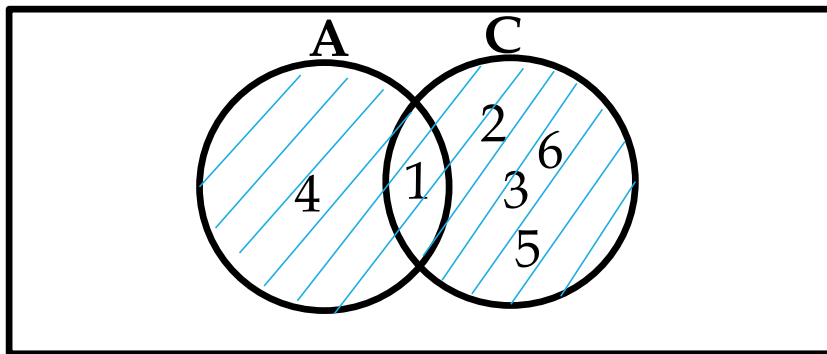
UNION AND INTERSECTION

❖ Union ($A \cup C$): The union of two events (A and C) represents all outcomes that are in either Event A, Event C, or both. In other words, it's the combination of all the elements in A and C.

$$A \cup C = \{1, 2, 3, 4, 5, 6\}$$

❖ Intersection ($A \cap C$): The intersection of two events (A and C) includes only the outcomes that are common to both events. It represents the overlap between A and C, meaning outcomes that occur in both events.

$$A \cap C = \{1\}$$



COMPLEMENTARY EVENTS

- ❖ The complement of an event A (denoted as A' or A^c) represents all outcomes in the sample space that are not in A .
- ❖ The sum of the probabilities of an event and its complement is always 1.
- ❖ $P(A) + P(A^c) = 1$
- ❖ Example: Event A : Rolling an even number on a die (2, 4, or 6).
- ❖ The complement of Event A (A^c) is rolling an odd number (1, 3, or 5).
- ❖ $P(A) = 3/6 = 0.5$ (since there are 3 even numbers)
- ❖ $P(A^c) = 3/6 = 0.5$ (since there are 3 odd numbers)
- ❖ $P(A) + P(A^c) = 0.5 + 0.5 = 1$

INDEPENDENT EVENTS

- ❖ Two events are said to be independent if the occurrence of one event does not affect the probability of the other event.
- ❖ In other words, knowing that one event occurred does not change the probability of the second event.
- ❖ Example:
- ❖ Event A: Rolling a 3 on a die.
- ❖ Event B: Flipping a heads on a coin.
- ❖ These two events are independent because rolling a die does not impact the outcome of flipping a coin, and vice versa.

RULE OF MULTIPLICATION (A “AND” B)

- ❖ For independent events - $P(A \cap B) = P(A) \times P(B)$
- ❖ Probability of rolling a 3 on a die ($P(A)$) = $1/6$
- ❖ Probability of flipping heads ($P(B)$) = $1/2$
- ❖ Since they are independent, the probability of both events occurring ($A \cap B$) is:
- ❖ $P(A \cap B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$

RULE OF MULTIPLICATION (A “AND” B)

- ❖ For dependent events (where one event affects the other), the formula is modified to:
- ❖ $P(A \cap B) = P(A) \times P(B|A)$
- ❖ Where $P(B|A)$ is the conditional probability of B given that A has occurred.
- ❖ Example: There are 10 candies in the jar: 5 Green, 2 Yellow, 2 Orange and 1 Red
- ❖ Question: If I pick 2 random candies, what is the probability of getting both Yellow? (Note: These are not independent events)

RULE OF MULTIPLICATION (A “AND” B)

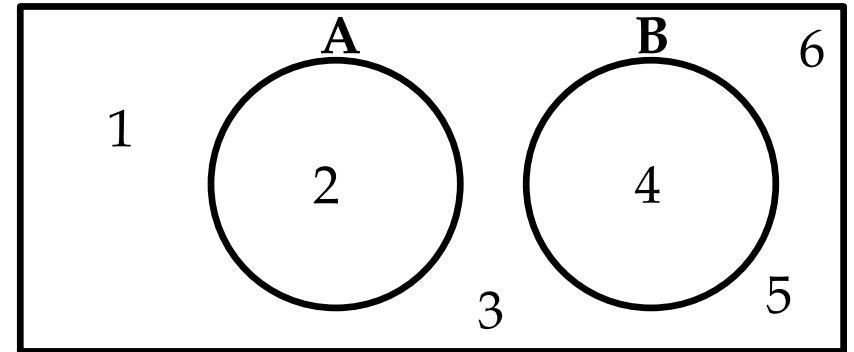
- ❖ First Draw (Without Replacement)
- ❖ Total Candies = 10
- ❖ Probability of first Yellow Candy ($P(Y1) = 2/10 = 0.2$)
- ❖ Second Draw (After First Yellow Candy). There are now: 1 Yellow Candy left and 9 Total Candies left
- ❖ Probability of second Yellow Candy ($P(Y2|Y1) = 1/9 = 0.111$)
- ❖ Step 3: Applying the Multiplication Rule for dependent event:
- ❖ $P(\text{both Yellow}) = P(Y1) \times P(Y2|Y1)$
- ❖ $P(\text{both Yellow}) = \left(\frac{2}{10}\right) \times \left(\frac{1}{9}\right) = \frac{2}{90} = 0.022$

LAW OF ADDITION (A “OR” B)

- ❖ The Law of Addition helps to find the probability that either of two events (A or B) will happen.
- ❖ For mutually exclusive events (events that cannot happen together), the probability of either event occurring is simply the sum of the individual probabilities.
- ❖ $P(A \cup B) = P(A) + P(B)$ (if A and B are mutually exclusive)
- ❖ For non-mutually exclusive events (events that can happen together), you subtract the probability of the intersection to avoid double counting.
- ❖ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

LAW OF ADDITION (A “OR” B)

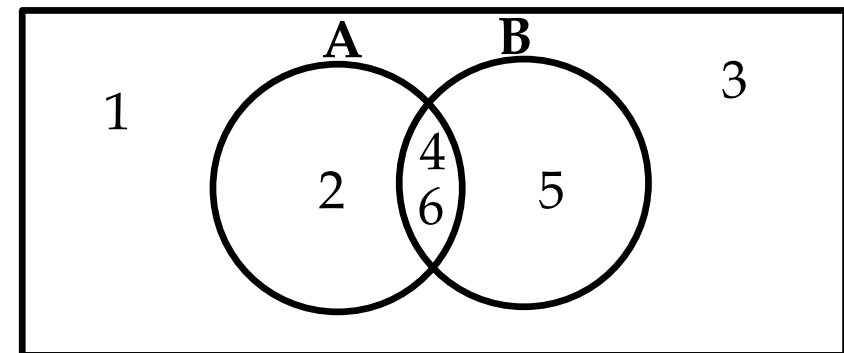
- ❖ Example 1: Mutually Exclusive Events
- ❖ Event A: Rolling a 2 on a die.
- ❖ Event B: Rolling a 4 on a die.
- ❖ Since you cannot roll both a 2 and a 4 at the same time, the events are mutually exclusive. Therefore:
- ❖ $P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$



LAW OF ADDITION (A “OR” B)

❖ Example 2: Non-Mutually Exclusive Events

- ❖ Event A: Rolling an even number on a die.
- ❖ Event B: Rolling a number greater than 3.
- ❖ In this case, the events are not mutually exclusive because some outcomes are common (rolling a 4 or 6). Therefore:
- ❖ $P(A)=3/6=0.5$ (Even numbers: 2, 4, 6)
- ❖ $P(B)=3/6=0.5$ (Numbers > 3 : 4, 5, 6)
- ❖ $P(A \cap B)=2/6=0.333$ (Common outcomes: 4, 6)
- ❖ Now applying the Law of Addition:
 - ❖ $P(A \cup B)=P(A)+P(B)-P(A \cap B)=0.5+0.5-0.333=0.667$



SUMMARY

Rule Type	Formula	Description
Multiplication Rule (Independent Events)	$P(A \cap B) = P(A) \times P(B)$	For independent events, where the outcome of one event does not affect the other.
Multiplication Rule (Dependent Events)	$P(A \cap B) = P(A) \times P(B A)$	For dependent events, where the outcome of one event affects the outcome of another.
Addition Rule (Mutually Exclusive Events)	$P(A \cup B) = P(A) + P(B)$	For mutually exclusive events, where both events cannot occur at the same time.
Addition Rule (Non-Mutually Exclusive Events)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	For non-mutually exclusive events, where both events can occur together (overlap between events).
Complementary Rule	$P(A') = 1 - P(A)$	The probability of the complement of event A, which is 1 minus the probability of A occurring.

CONDITIONAL PROBABILITY

- ❖ Conditional probability is the probability of an event A occurring, given that event B has already occurred.
- ❖ Formula:
- ❖
$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$
- ❖ Where $P(A|B)$ is the probability of A given B, and $P(A \cap B)$ is the probability of both events occurring together.
- ❖ Key Insight:
- ❖ Conditional probability modifies the original probability by limiting the sample space to only those outcomes where event B occurs.

CONDITIONAL PROBABILITY

- ❖ A deck of 52 cards. What is the probability of drawing an Ace given that the card drawn is a Spade?
- ❖ Solution:
- ❖ There are 13 Spades, and only 1 of them is an Ace.
- ❖ $P(\text{Ace} \mid \text{Spade}) = 1/13$
- ❖ Explanation:
- ❖ The probability is conditioned on the card being a Spade, reducing the sample space from 52 cards to 13.

FACTORIAL (REFRESHER)

- ❖ The factorial of a non-negative integer n , denoted as $n!$, is the product of all positive integers less than or equal to n .
- ❖ Formula:
- ❖ $n! = n \times (n - 1) \times (n - 2) \times \cdots \times 1$
- ❖ Special Case:
- ❖ $0! = 1$
- ❖ Examples:
- ❖ $4! = 4 \times 3 \times 2 \times 1 = 24$
- ❖ $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

PERMUTATIONS

- ❖ A permutation refers to an arrangement of objects in a specific order.
- ❖ Formula: The number of ways to arrange r objects from n distinct objects:
- ❖
$$P(n, r) = \frac{n!}{(n-r)!}$$
- ❖ Key Concept: Order matters in permutations.
- ❖ Example: How many ways can you arrange 3 books from a collection of 5 books?
- ❖
$$P(5, 3) = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3}{1} = 60$$
- ❖ There are 60 possible arrangements.

PERMUTATION WITH REPETITION

- ❖ Permutations where objects can be repeated (e.g., digits in a PIN code).
- ❖ Formula:
- ❖ If there are n objects and each can be chosen multiple times, the total number of permutations is: n^r
- ❖ Example: How many 4-digit PIN codes can be formed using 6 digits (0-5)?
- ❖ $6^4 = 6 \times 6 \times 6 \times 6 = 1296$

COMBINATIONS

- ❖ A combination refers to a selection of objects without regard to the order.
- ❖ Formula:
- ❖ The number of ways to choose r objects from n distinct objects:
- ❖
$$C(n, r) = \frac{n!}{r!(n-r)!}$$
- ❖ Key Concept: Order does not matter in combinations.
- ❖ Example: How many ways can you choose 3 fruits from a basket of 5 fruits?
- ❖
$$C(5, 3) = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5 \times 4}{2 \times 1} = 10$$
- ❖ There are 10 possible combinations.

COMBINATION WITH REPETITION

- ❖ Combinations where objects can be selected more than once (e.g., scoops of ice cream flavors).
- ❖ Formula: The number of combinations when repetition is allowed:
- ❖
$$C(n + r - 1, r) = \frac{(n+r-1)!}{r!(n-1)!}$$
- ❖ Example: How many ways can you choose 3 scoops of ice cream from 5 flavors, with repetitions allowed?
- ❖
$$C(5 + 3 - 1, 3) = C(7, 3) = \frac{7!}{3!4!} = 35$$

PERMUTATIONS AND COMBINATIONS (SUMMARY)

	With repetition	Without repetition
❖ Permutation: Order is important. MS Excel Formula >>	n^r PERMUTATIONA(n,r)	$n_{P_r} = \frac{n!}{(n-r)!}$ PERMUT(n,r)
❖ Combination: Order is NOT important. MS Excel Formula >>	$\frac{(r+n-1)!}{r!(n-1)!}$ COMBINA(n,r)	$n_{C_r} = \frac{n!}{(n-r)! r!}$ COMBIN(n,r)

EXCEL DEMO — PROBABILITY RELATED FUNCTIONS

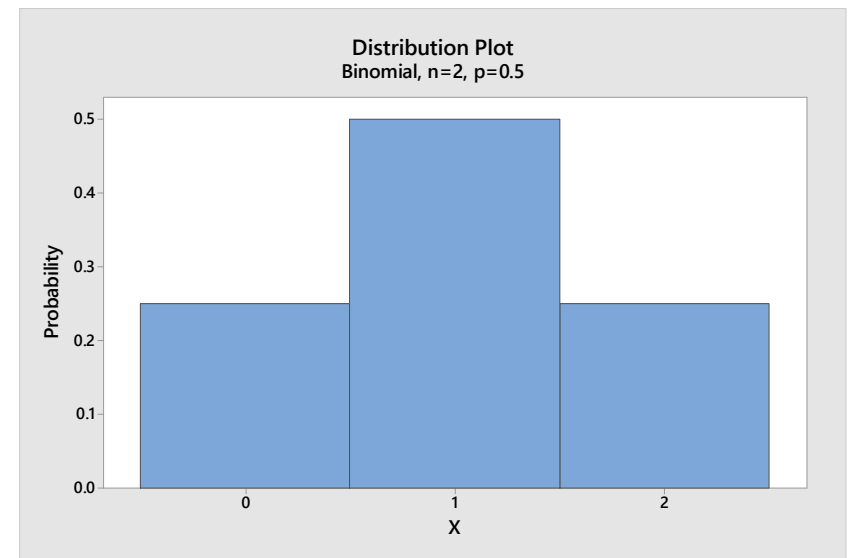
Function	Description	Example	Output
RAND()	Generates a random number between 0 and 1	=RAND()	0.423 (varies)
RANDBETWEEN	Generates a random integer between two bounds	=RANDBETWEEN(1, 6)	4 (varies)
FACT	Returns the factorial of a number	=FACT(5)	120
PERMUT	Returns the number of permutations (order matters)	=PERMUT(5, 3)	60
PERMUTATIONA	Returns the number of permutations allowing repetitions	=PERMUTATIONA(5, 3)	125
COMBIN	Returns the number of combinations (order does not matter)	=COMBIN(5, 3)	10
COMBINA	Returns the number of combinations allowing repetitions	=COMBINA(5, 3)	35
ROUND	Rounds a number to a specified number of digits	=ROUND(0.123456, 2)	0.12

COMMON DISTRIBUTIONS

- ❖ A statistical distribution describes how values or outcomes of a random variable are distributed across a range of possibilities.
- ❖ In simple terms, it tells us the probability of different outcomes in a random experiment.
- ❖ Understanding distributions helps in predicting the behavior of random variables in real-life scenarios.
- ❖ Different distributions are used to model different types of data (e.g., discrete or continuous).
 - ❖ Discrete Distributions (e.g., Binomial, Poisson): Used for countable data.
 - ❖ Continuous Distributions (e.g., Normal, Chi-Square): Used for data that can take any value within a range.

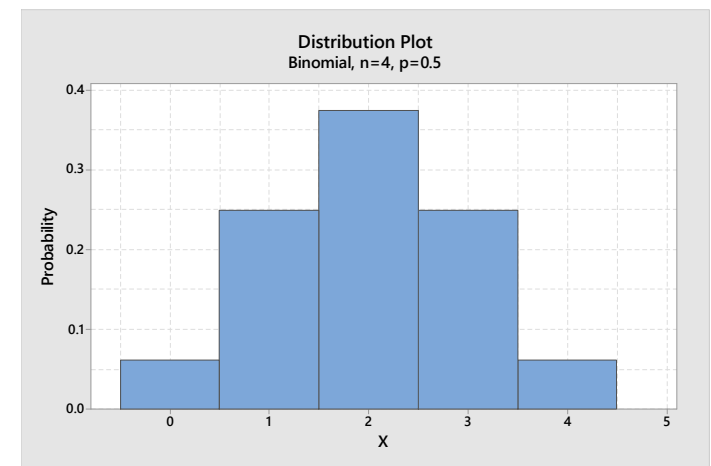
FLIPPING 2 COINS

- ❖ When flipping 2 coins, each coin can either land on Heads (H) or Tails (T).
- ❖ The sample space for 2 coins is: {HH, HT, TH, TT}.
- ❖ Possible Outcomes:
- ❖ 0 Heads: TT (1 outcome)
- ❖ 1 Head: HT, TH (2 outcomes)
- ❖ 2 Heads: HH (1 outcome)



FLIPPING 4 COINS

- ❖ When flipping 4 coins, the outcomes are more diverse. Each coin still has two possibilities: H or T.
- ❖ The sample space consists of 16 possible outcomes (e.g., {HHHH, HHHT, HHTT, ...}).
- ❖ Possible Outcomes:
 - ❖ 0 Heads: TTTT (1 outcome)
 - ❖ 1 Head: HTTT, THTT, TTHT, TTTH (4 outcomes)
 - ❖ 2 Heads: HHTT, HTHT, HTTH, THHT, THTH, TTHT (6 outcomes)
 - ❖ 3 Heads: HHHT, HHTH, HTHH, THHH (4 outcomes)
 - ❖ 4 Heads: HHHH (1 outcome)



BINOMIAL DISTRIBUTION

- ❖ Flipping a coin 4 times and calculating probabilities is straightforward because each event is independent, and the probability of heads is 0.5. We can calculate probabilities easily by listing all possible outcomes (e.g., getting 2 heads).
- ❖ In real-world scenarios, probabilities may not be as simple. Example: A manufacturer has a 12% defect rate. The buyer tests 20 random pieces and will only accept the supplier if there are 2 or fewer defectives.
- ❖ Calculating the probability of acceptance becomes more complex with varying defect rates and sample sizes.
- ❖ Solution: **Binomial Distribution**

BINOMIAL DISTRIBUTION

- ❖ A Binomial Distribution models the number of successes in a fixed number of independent trials, where each trial has two possible outcomes (success or failure).
- ❖ The distribution is defined by:
 - ❖ n : Number of trials.
 - ❖ p : Probability of success on each trial.

BINOMIAL DISTRIBUTION

Properties of Binomial Distribution:

- ❖ **Binary outcome:** Each trial has only two possible outcomes (e.g., success/failure, defective/non-defective).
- ❖ **Fixed number of trials:** The number of trials, n , is predetermined.
- ❖ **Constant probability:** The probability of success, p , remains the same for each trial.
- ❖ **Independence:** Each trial is independent of the others.

BINOMIAL DISTRIBUTION

❖ Binomial Probability Formula:

❖ $P(X = x) = {}^nC_x \cdot p^x \cdot (1 - p)^{n-x}$

❖ Where:

- ❖ $P(X = x)$ is the probability of getting exactly x successes.
- ❖ n is the total number of trials.
- ❖ p is the probability of success on each trial.
- ❖ x is the number of successes you want to calculate the probability for.

❖ Factorial Explanation:

❖ $({}^nC_x)$ is the number of ways to choose x successes from n trials and is calculated using factorials: ${}^nC_x = \frac{n!}{x!(n-x)!}$

EXAMPLE: FLIPPING 4 COINS

- ❖ Suppose you flip 4 coins, and you want to know the probability of getting exactly 1 head.
- ❖ Solution:
- ❖ $P(X = 1) = (4C1) \cdot (0.5)^1 \cdot (1 - 0.5)^{4-1}$
- ❖ $P(X = 1) = \left(\frac{4!}{1!3!}\right) \cdot (0.5)^1 \cdot (0.5)^3$
- ❖ Interpretation:
- ❖ The probability of getting exactly 1 head when flipping 4 coins is 25%.

EXAMPLE: DEFECT RATE OF 12%

- ❖ Problem: A manufacturer has a 12% defect rate. The buyer will test 20 pieces and will only accept the supplier if 2 or fewer pieces are defective.
- ❖ Variables:
 - ❖ $n = 20$: Number of pieces tested.
 - ❖ $p = 0.12$: Probability of a defective piece.
 - ❖ $x = 0, 1, 2$: The acceptable number of defective pieces.
- ❖ Solution Using Binomial Formula:
 - ❖ $P(X=0 \text{ or } 1 \text{ or } 2) = P(X=0) + P(X=1) + P(X=2)$
 - ❖ Each of these can be calculated using the binomial formula to find the overall probability of acceptance.

EXAMPLE: DEFECT RATE OF 12%

❖ $p = 0.12, n = 20, x = 0, 1, 2$ $P(x) = \frac{n!}{x! (n - x)!} \cdot p^x \cdot (1 - p)^{n-x}$

$$P(0) = \frac{20!}{0! (20 - 0)!} \cdot 0.12^0 \cdot (1 - 0.12)^{20-0}$$

$$P(1) = \frac{20!}{1! (20 - 1)!} \cdot 0.12^1 \cdot (1 - 0.12)^{20-1}$$

$$P(2) = \frac{20!}{2! (20 - 2)!} \cdot 0.12^2 \cdot (1 - 0.12)^{20-2}$$

$$P(0,1,2) = 0.077563 + 0.211535 + 0.274034 = 0.563132$$

MEAN OF BINOMIAL DISTRIBUTION

- ❖ Mean Formula: $\mu = n \cdot p$
- ❖ Where:
 - ❖ μ is the expected number of successes.
 - ❖ n is the number of trials.
 - ❖ p is the probability of success in each trial.
- ❖ Mean (μ) represents the expected number of successes in n independent trials.
- ❖ Example: If you flip a coin 10 times ($n=10$) and the probability of heads (p) is 0.5, the expected number of heads:
- ❖ $\mu = 10 \times 0.5 = 5$
- ❖ This means, on average, you expect 5 heads out of 10 flips.

VARIANCE OF BINOMIAL DISTRIBUTION

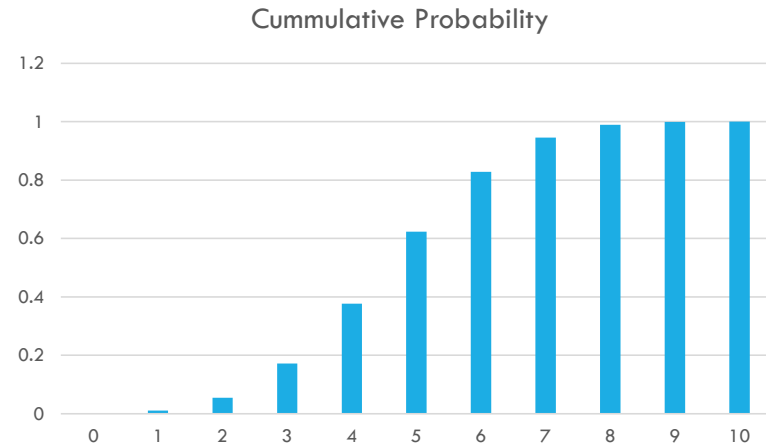
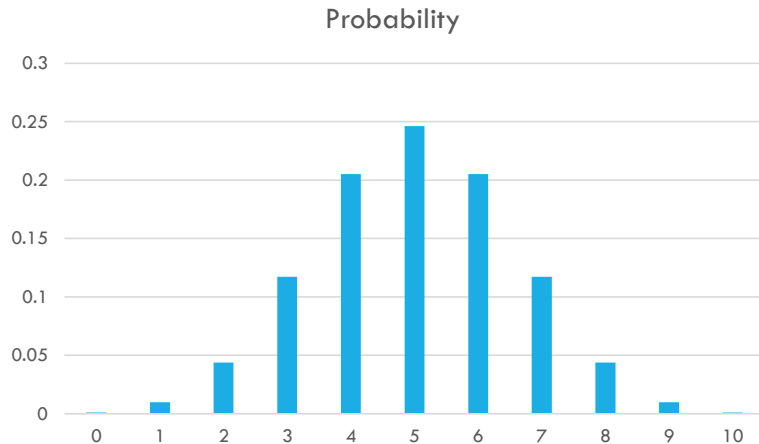
- ❖ Variance Formula: $\sigma^2 = n \cdot p \cdot (1 - p)$
- ❖ Where:
 - ❖ σ^2 is the variance.
 - ❖ n is the number of trials.
 - ❖ p is the probability of success.
- ❖ Variance (σ^2) represents how much the number of successes is expected to vary around the mean.
- ❖ Example: For 10 coin flips, the variance of the number of heads:
- ❖ $\sigma^2 = 10 \cdot 0.5 \cdot (1 - 0.5) = 2.5$
- ❖ The standard deviation is simply the square root of the variance.

BINOM.DIST() FUNCTION

- ❖ Purpose: Calculates the probability of a specific number of successes in a binomial experiment.
- ❖ Syntax: =BINOM.DIST(successes, trials, probability, cumulative)
 - ❖ trials: Total number of trials.
 - ❖ successes: Number of successes you want to calculate the probability for.
 - ❖ probability: Probability of success in each trial.
 - ❖ cumulative: TRUE for cumulative probability, FALSE for exact probability.
- ❖ Example: A coin is flipped 10 times. What is the probability of getting exactly 5 heads?
- ❖ =BINOM.DIST(5, 10, 0.5, FALSE) >>> Output: 0.246

BINOM.DIST() FUNCTION

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- ❖ `=BINOM.DIST(5, 10, 0.5, FALSE)` >>> Output: 0.246



BINOM.DIST.RANGE() FUNCTION

- ❖ Purpose: Returns the probability of a range of successes in a binomial experiment.
- ❖ Syntax: `=BINOM.DIST.RANGE(trials, probability, lower_limit, [upper_limit])`
 - ❖ trials: Total number of trials.
 - ❖ probability: Probability of success in each trial.
 - ❖ lower_limit: Lower bound of the number of successes.
 - ❖ [upper_limit]: (Optional) Upper bound of the number of successes.
- ❖ Example: What is the probability of getting between 3 and 5 heads (inclusive) in 10 coin flips?
- ❖ `=BINOM.DIST.RANGE(10, 0.5, 3, 5)` >>> Output: 0.568

BINOM.INV() FUNCTION

- ❖ Purpose: Returns the smallest number of successes for which the cumulative binomial distribution is greater than or equal to a specified criterion.
- ❖ Syntax: =BINOM.INV(trials, probability, alpha)
 - ❖ trials: Total number of trials.
 - ❖ probability: Probability of success in each trial.
 - ❖ alpha: Cumulative probability threshold.
- ❖ Example: In 10 coin flips, what is the smallest number of heads for which the cumulative probability is greater than or equal to 0.8?
- ❖ =BINOM.INV(10, 0.5, 0.8) >>> Output: 7 (At least 7 heads are needed to reach or exceed a cumulative probability of 0.8).

QUESTION 1: DEFECT RATE IN A BATCH

- ❖ Scenario: A manufacturer produces 50 electronic components in a batch. Historically, the defect rate is 5%. The quality control team inspects the batch and will reject it if more than 3 defective components are found. What is the probability that the batch will be accepted?
- ❖ Solution:
 - ❖ $n = 50$ (number of trials).
 - ❖ $p = 0.05$ (probability of a defective component).
 - ❖ Acceptable defective components: $x = 0, 1, 2, 3$.
- ❖ The cumulative probability: `=BINOM.DIST.RANGE(50, 0.05, 0, 3)`
- ❖ Output: The probability of acceptance is 0.789, or 78.9%.

QUESTION 2: LOAN DEFAULT RISK

- ❖ Scenario: A bank issues 20 loans. Based on historical data, the probability of a loan defaulting is 10%. What is the probability that, at most, 1 loan will default?
- ❖ Solution:
 - ❖ $n = 20$ (number of trials).
 - ❖ $p = 0.10$ (probability of default).
 - ❖ $x = 0, 1$ (at most 1 default).
- ❖ The cumulative probability: $=\text{BINOM.DIST.RANGE}(20, 0.10, 0, 1)$
- ❖ Output: The probability that at most 1 loan default is 0.3917, or 39.17%.

QUESTION 2: LOAN DEFAULT RISK

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 - ❖ $x = 0, 1$ (at most 1 default).
- ❖ The cumulative probability: $=\text{BINOM.DIST.RANGE}(20, 0.10, 0, 1)$
- ❖ Or $=\text{BINOM.DIST}(1, 20, 0.1, \text{TRUE})$
- ❖ Output: The probability that at most 1 loan default is 0.3917, or 39.17%.

QUESTION 3: DEFECT RATE IN A BATCH

- ❖ Scenario: As a seller, you want to be 90% confident that a batch will be accepted by the client after they inspect 50 pieces. Historically, the defect rate is 5%. What is the maximum number of defective pieces you should allow (quote to the client) in the batch to ensure a 90% probability that the batch will be accepted?
- ❖ Solution:
 - ❖ $n = 50$ (number of trials).
 - ❖ $p = 0.05$ (probability of a defective component).
 - ❖ $\text{Alpha} = 0.90$.
- ❖ The cumulative probability: $=\text{BINOM.INV}(50, 0.05, 0.90)$
 - ❖ Output: The result is 5, meaning you can quote up to 5 defective pieces in the batch of 50 pieces to be 90% confident that the client will accept the batch.

POISSON DISTRIBUTION

- ❖ Difference from Binomial Distribution:
- ❖ Binomial Distribution: Used when the number of trials n is fixed, and we want to know the probability of getting exactly k successes.
- ❖ Poisson Distribution: Used when we know the average number of events happening in a fixed interval of time/space and want to find the probability of k events occurring.
- ❖ When to Use Poisson:
 - ❖ When events happen independently.
 - ❖ The average rate of occurrence is known.
 - ❖ The number of trials is large, but the probability of occurrence in each trial is small.

POISSON DISTRIBUTION

- ❖ Key Properties:
- ❖ Poisson is defined by a single parameter: λ (mean number of events per unit time).
- ❖ *** A Poisson distribution's mean and variance are equal to λ .
- ❖ Example:
- ❖ If 4 cars pass by a toll booth every minute on average, $\lambda=4$.

POISSON DISTRIBUTION

❖ Poisson Probability Mass Function (PMF):

$$\text{❖ } P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

❖ Where:

- ❖ $P(X=k)$: Probability of observing exactly k events.
- ❖ λ : Average number of events per interval.
- ❖ e : Euler's number (approx. 2.71828).
- ❖ k : Number of events.
- ❖ $k!$: Factorial of k .

POISSON DISTRIBUTION - EXAMPLE

- ❖ On a booking counter, an average of 3.6 people come every 10 minutes on weekends. What is the probability of getting exactly 7 people in 10 minutes?
- ❖ Given:
 - ❖ $\lambda=3.6$ (average number of people per 10 minutes)
 - ❖ $k=7$ (number of people we're interested in)
- ❖ $P(X = 7) = \frac{3.6^7 \cdot e^{-3.6}}{7!}$
- ❖ $P(X=7) \approx 7836.41 \times 0.02732 / 5040 = 0.0424$

POISSON.DIST() FUNCTION

- ❖ Purpose: Calculates the probability of observing a specific number of events in a given time interval.
- ❖ Syntax: `=POISSON.DIST(x, mean, cumulative)`
 - ❖ x: Number of events.
 - ❖ mean: Expected number of events in the interval.
 - ❖ cumulative:
 - ❖ TRUE: Returns cumulative probability ($P(X \leq x)$).
 - ❖ FALSE: Returns probability mass function ($P(X = x)$).
- ❖ Example: A call center receives 3.6 calls per hour on average. What is the probability that the center receives exactly 5 calls in an hour?
- ❖ `=POISSON.DIST(5, 3.6, FALSE)` >>> Output: 0.1377 (There is a 13.77% chance of receiving exactly 5 calls in an hour).

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POISSON.DIST() FUNCTION

Distribution	Functions in Excel	Reason
Binomial	BINOM.DIST, BINOM.DIST.RANGE, BINOM.INV	Different types of binomial problems: exact probability, cumulative range, and inverse calculations.
Poisson	POISSON.DIST	Single function suffices because it handles both exact and cumulative probabilities.

PMF VS PDF: KEY DIFFERENCES

❖ PMF (Probability Mass Function):

- ❖ A function that gives the probability of a discrete random variable taking a particular value.
- ❖ Used for Discrete Distributions (e.g., Binomial, Poisson).
- ❖ The probability is assigned to each distinct outcome.
- ❖ The PMF is represented as bars at distinct values of the variable (e.g., Binomial or Poisson).
- ❖ Example: Probability of getting exactly 3 heads when flipping 5 coins.

❖ PDF (Probability Density Function):

- ❖ A function that gives the probability density of a continuous random variable at a particular value.
- ❖ Used for Continuous Distributions (e.g., Normal, Exponential).
- ❖ Probability for a single point is 0, but the area under the curve in an interval gives the probability of that interval.
- ❖ The PDF is represented as a smooth curve (e.g., Normal distribution curve).
- ❖ Example: Probability of a person's height falling between 160 cm and 170 cm.

PMF VS PDF: KEY DIFFERENCES

❖ PMF (Discrete Random Variables):

- ❖ Used for discrete random variables.
- ❖ Gives exact probabilities.
- ❖ Sum of all probabilities is 1.
- ❖ Example: Poisson distribution, Binomial distribution.

❖ PDF (Continuous Random Variables):

- ❖ Used for continuous random variables.
- ❖ Provides density, not exact probabilities. To get a probability, you integrate over an interval.
- ❖ Total area under the curve is 1.
- ❖ Example: Normal distribution, Exponential distribution.

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

- ❖ The CDF gives the cumulative probability of a random variable being less than or equal to a certain value.
- ❖ For Discrete Variables (PMF):
 - ❖ CDF is the sum of the PMF values up to a certain point.
 - ❖ CDF of a binomial distribution up to $k = 3$ is the sum of the PMF values for $k = 0, 1, 2, 3$.
- ❖ For Continuous Variables (PDF):
 - ❖ CDF is the integral of the PDF up to a certain point.
 - ❖ CDF for a normal distribution tells us the probability that a value will be below a given point.

NORMAL DISTRIBUTION

- ❖ The Normal distribution, also known as the **Gaussian distribution**, is a **continuous probability distribution** that is symmetric about the mean.
- ❖ Shape: Bell-shaped curve (unimodal and symmetric).
- ❖ Key Characteristics:
 - ❖ Mean (μ), Standard Deviation (σ)
 - ❖ Mean = Median = Mode (symmetric)
 - ❖ Total area under the curve = 1

PROBABILITY DENSITY FUNCTION (PDF)

❖ Formula:

$$P(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

❖ Where:

- ❖ μ = mean
- ❖ σ = standard deviation
- ❖ x = any value on the x-axis

❖ Interpretation:

- ❖ The height of the PDF at a point represents the probability density.
- ❖ To find the probability of an interval (e.g., $x_1 \leq X \leq x_2$), you calculate the area under the curve between those points.

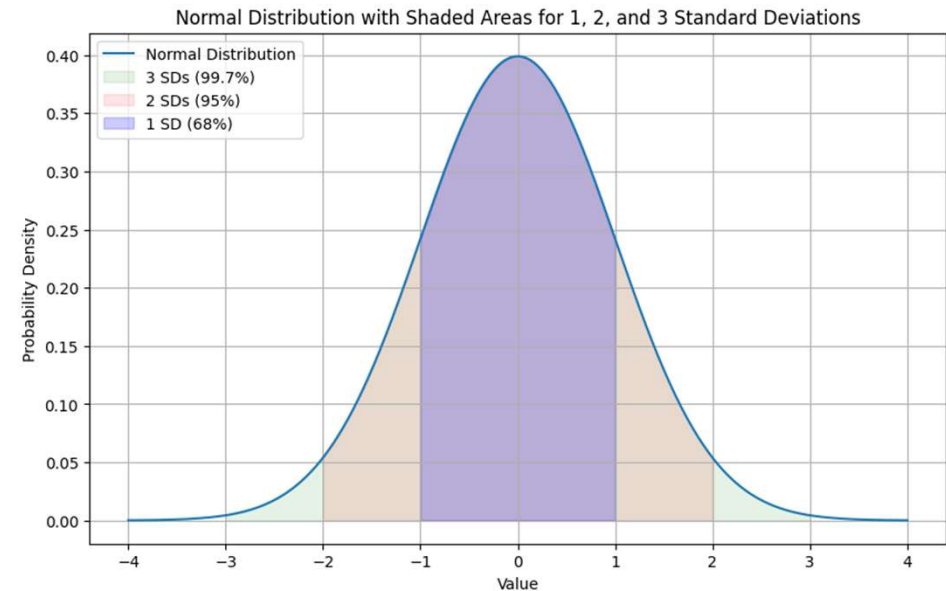
NORMAL DISTRIBUTION - PROPERTIES

- ❖ **Symmetry:** The curve is perfectly symmetric about the mean.
- ❖ **Mean, Median, Mode:** All are equal, lying at the center of the distribution.
- ❖ **68-95-99.7 Rule (Empirical Rule):**
 - ❖ 68% of data lies within 1σ of the mean.
 - ❖ 95% of data lies within 2σ of the mean.
 - ❖ 99.7% of data lies within 3σ of the mean.
- ❖ **Continuous:** There are infinite possible values within the distribution. The probability of a specific outcome is zero.

NORMAL DISTRIBUTION - PROPERTIES

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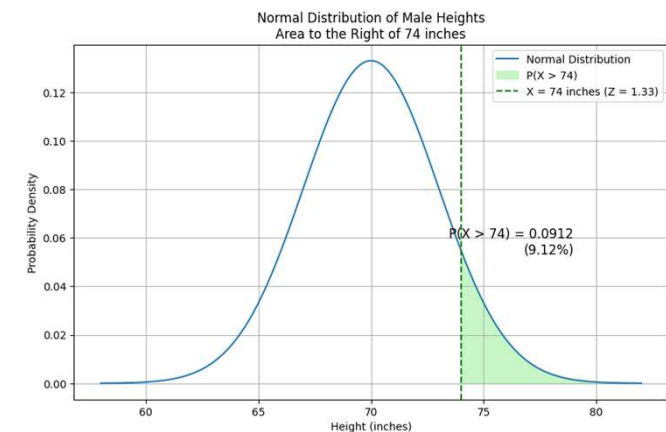


STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

- ❖ A normal distribution with a mean of 0 and a standard deviation of 1.
- ❖ Z-Score Formula:
- ❖ $Z = \frac{X - \mu}{\sigma}$
- ❖ Interpretation: The Z-score tells us how many standard deviations away a value (X) is from the mean.
- ❖ Example: A Z-score of 2 means the value is 2 standard deviations above the mean.
- ❖ Standard Normal Table: Used to find probabilities for any Z-score.

STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

- ❖ Heights of adult males follow a normal distribution with a mean of 70 inches and a standard deviation of 3 inches.
- ❖ Question: What is the probability of selecting a male taller than 74 inches?
- ❖ $Z = \frac{74-70}{3} = 1.33$
- ❖ Find Probability from Z-Table:
- ❖ $P(Z > 1.33) = 0.0918$
- ❖ Answer: There is a 9.18% probability that a randomly selected male is taller than 74 inches.



STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

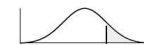
❖ Find Probability from Z-Table:

❖ $P(Z > 1.33) = 0.0918$

z	0.00	0.01	0.02	0.03	0.04
0	0.5000	0.4960	0.4920	0.4880	0.4840
0.1	0.4602	0.4562	0.4522	0.4483	0.4443
0.2	0.4207	0.4168	0.4129	0.4090	0.4050
0.3	0.3821	0.3783	0.3745	0.3707	0.3669
0.4	0.3446	0.3409	0.3372	0.3336	0.3299
0.5	0.3085	0.3050	0.3015	0.2981	0.2946
0.6	0.2743	0.2709	0.2676	0.2643	0.2610
0.7	0.2420	0.2389	0.2358	0.2327	0.2296
0.8	0.2119	0.2090	0.2061	0.2033	0.2005
0.9	0.1841	0.1814	0.1788	0.1762	0.1735
1.0	0.1587	0.1562	0.1539	0.1515	0.1492
1.1	0.1357	0.1335	0.1314	0.1292	0.1271
1.2	0.1151	0.1131	0.1112	0.1093	0.1075
1.3	0.0968	0.0951	0.0934	0.0918	0.0901
1.4	0.0808	0.0793	0.0778	0.0764	0.0749

Normal Distribution Tables

Table 1: Areas Under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4050	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2610	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1735	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002

EXAMPLE 2: PROBABILITY LESS THAN A VALUE

- ❖ What is the probability of selecting a male shorter than 65 inches, given that the mean height is 70 inches and the standard deviation is 3 inches?
- ❖ $Z1 = \frac{65-70}{3} = -1.67$ (for 65 inches)

STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

❖ Find Probability from Z-Table:

❖ $P(Z > 1.67) = 0.0475$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.0
0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4
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0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0

Normal Distribution Tables

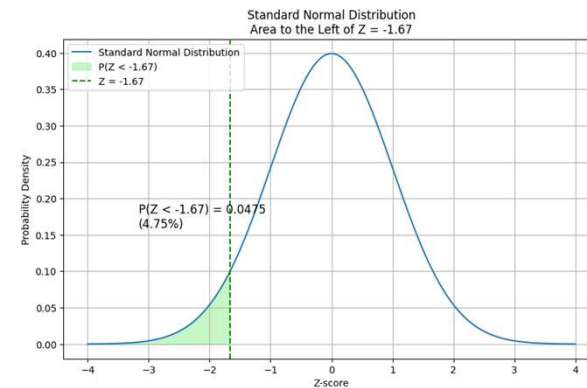
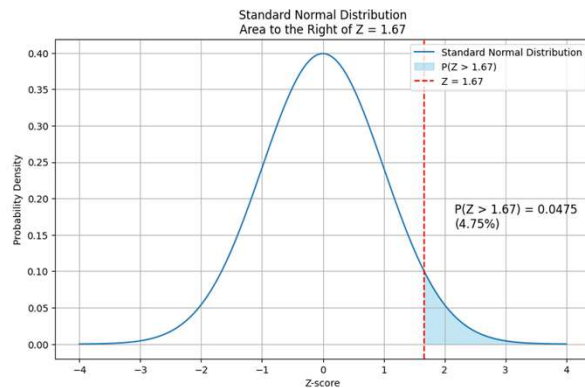
Table 1: Areas Under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002

EXAMPLE 2: PROBABILITY LESS THAN A VALUE

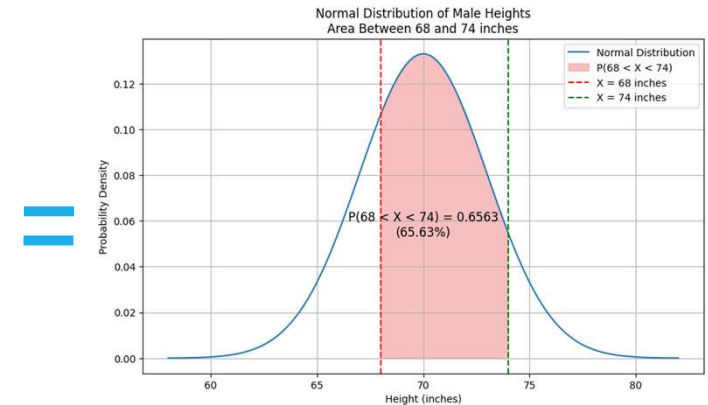
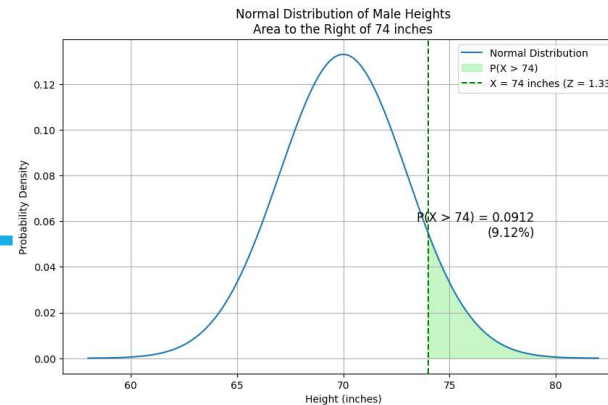
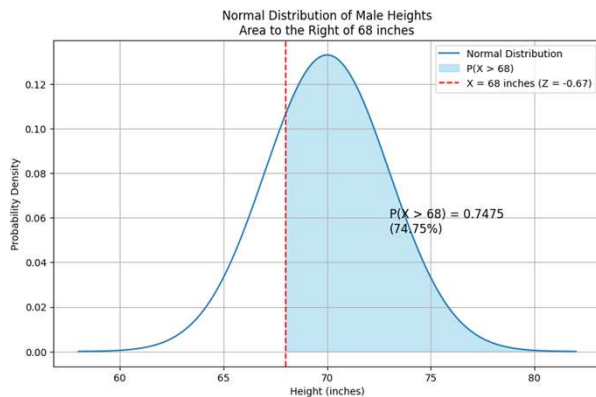
- ❖ What is the probability of selecting a male shorter than 65 inches, given that the mean height is 70 inches and the standard deviation is 3 inches?
- ❖ $Z1 = \frac{65-70}{3} = -1.67$ (for 65 inches)



Answer: The probability of selecting a male shorter than 65 inches is **4.75%**.

EXAMPLE 3: PROBABILITY BETWEEN TWO HEIGHTS

- ❖ What is the probability of selecting a male between 68 and 74 inches tall, given that the mean height is 70 inches and the standard deviation is 3 inches?
- ❖ $Z_1 = \frac{68-7}{3} = -0.67$ (for 68 inches)
- ❖ $Z_2 = \frac{74-7}{3} = 1.33$ (for 74 inches)

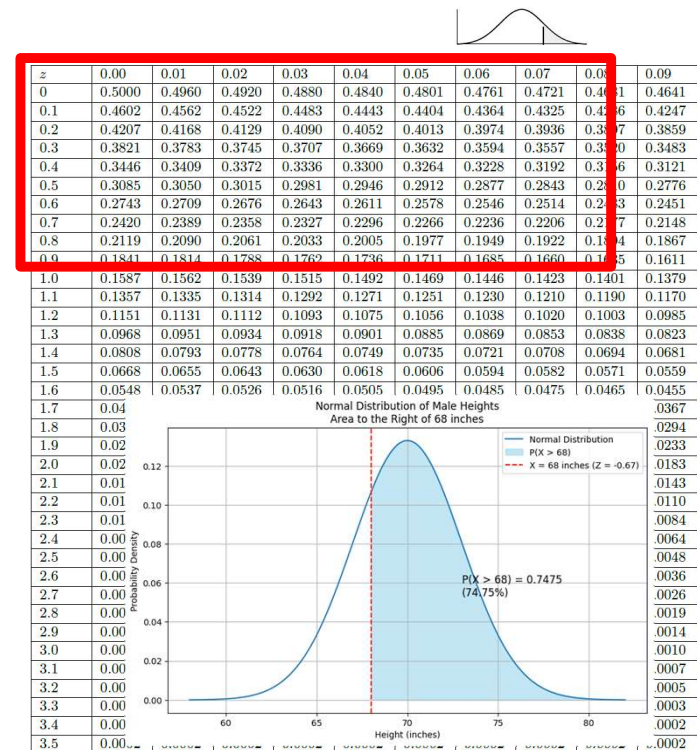


EXAMPLE 3: PROBABILITY BETWEEN TWO HEIGHTS

Normal Distribution Tables

Table 1: Areas Under the Standard Normal Curve

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07
0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660



$$P(X > 68) = 1.00 - 0.2514 = 0.7476$$

STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

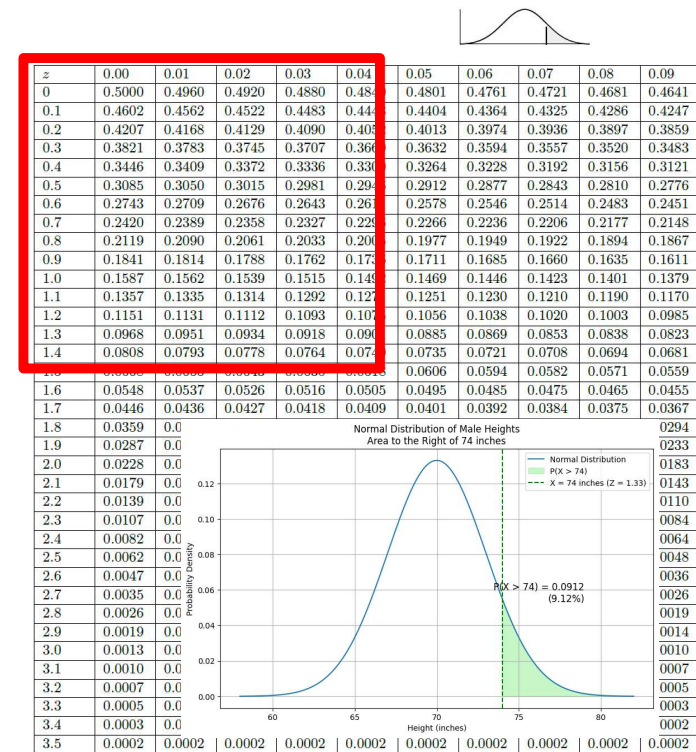
❖ Find Probability from Z-Table:

❖ $P(Z > 1.33) = 0.0918$

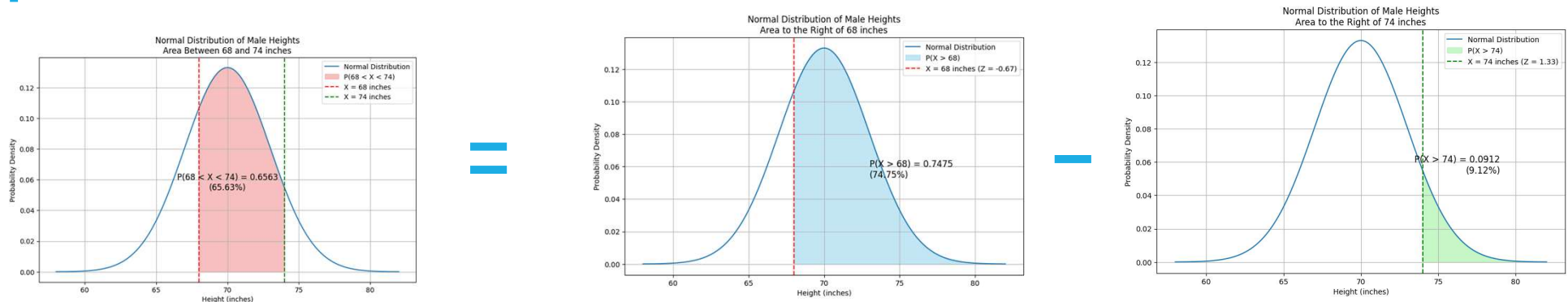
z	0.00	0.01	0.02	0.03	0.04
0	0.5000	0.4960	0.4920	0.4880	0.4840
0.1	0.4602	0.4562	0.4522	0.4483	0.4443
0.2	0.4207	0.4168	0.4129	0.4090	0.4050
0.3	0.3821	0.3783	0.3745	0.3707	0.3669
0.4	0.3446	0.3409	0.3372	0.3336	0.3299
0.5	0.3085	0.3050	0.3015	0.2981	0.2946
0.6	0.2743	0.2709	0.2676	0.2643	0.2610
0.7	0.2420	0.2389	0.2358	0.2327	0.2296
0.8	0.2119	0.2090	0.2061	0.2033	0.2005
0.9	0.1841	0.1814	0.1788	0.1762	0.1735
1.0	0.1587	0.1562	0.1539	0.1515	0.1492
1.1	0.1357	0.1335	0.1314	0.1292	0.1271
1.2	0.1151	0.1131	0.1112	0.1093	0.1075
1.3	0.0968	0.0951	0.0934	0.0918	0.0901
1.4	0.0808	0.0793	0.0778	0.0764	0.0750

Normal Distribution Tables

Table 1: Areas Under the Standard Normal Curve



EXAMPLE 2: PROBABILITY BETWEEN TWO HEIGHTS



$$P(68 < X < 74) = 0.7475 - 0.0912 = 0.6563$$

Answer: The probability of selecting a male between 68 and 74 inches tall is **65.63%**.

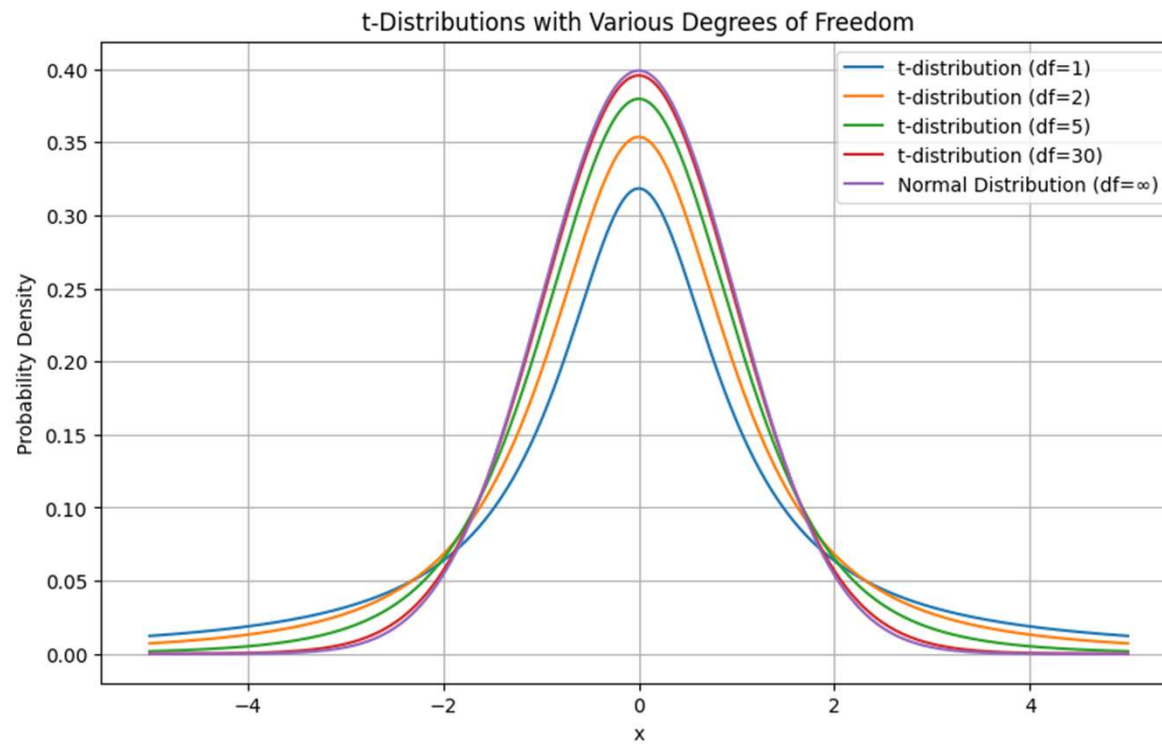
STUDENT'S T-DISTRIBUTION

- ❖ A probability distribution used when the sample size is small, and the population standard deviation is unknown.
- ❖ It is similar to the normal distribution but has heavier tails, meaning it accounts for more variability in small samples.
- ❖ Why is it important?
- ❖ Used for hypothesis testing (t-tests) and confidence intervals, especially when dealing with smaller sample sizes (typically $n < 30$).

STUDENT'S T-DISTRIBUTION

- ❖ Key Characteristics:
- ❖ Symmetrical and bell-shaped like the normal distribution but with thicker tails.
- ❖ As the sample size increases, the t-distribution approaches the normal distribution.
- ❖ Depends on degrees of freedom (df), which is related to the sample size ($df = n - 1$ for a single sample).
- ❖ Degrees of Freedom (df):
 - ❖ Lower degrees of freedom mean wider, flatter distributions.
 - ❖ Higher degrees of freedom bring it closer to the standard normal distribution.

STUDENT'S T-DISTRIBUTION



STUDENT'S T-DISTRIBUTION

❖ Probability Density of t Distribution:

$$f(t|v) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi} \Gamma\left(\frac{v}{2}\right)} \left(1 + \frac{t^2}{v}\right)^{-\frac{v+1}{2}}$$

Where:

- t is the t-statistic,
- v is the degrees of freedom,
- Γ is the Gamma function.

STUDENT'S T-DISTRIBUTION

❖ t-Statistic Formula:

$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

❖ Where:

- ❖ $\bar{x}$ = sample mean
 - ❖ μ = population mean (hypothesized)
 - ❖ s = sample standard deviation
 - ❖ n = sample size
- ❖ Degrees of Freedom: $df=n-1$
- ❖ Used to find the critical value of t from the t-distribution table.

STUDENT'S T-DISTRIBUTION

❖ t-Statistic Formula:

$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

t Distribution Table

Table 2: Table of the Student's t-distribution



df	α						
	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.706	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646

STUDENT'S T-DISTRIBUTION (MEAN AND VARIANCE)

- ❖ Mean of t-Distribution:

- ❖ The mean of the t-distribution is 0, just like the standard normal distribution.

- ❖ Variance of t-Distribution:

- ❖ The variance of the t-distribution is given by:

- ❖ $Variance = \frac{\nu}{\nu-2}$

- ❖ Where:

- ❖ ν = degrees of freedom (df).

- ❖ ν (pronounced "nu") is the Greek letter nu.

- ❖ For $\nu > 2$ the variance is finite. (Variance can not be negative)

- ❖ As the degrees of freedom increase, the variance of the t-distribution approaches 1 (which is the variance of the normal distribution), meaning the variance gets closer to 1, and the mean remains at 0.

CHI-SQUARE DISTRIBUTION

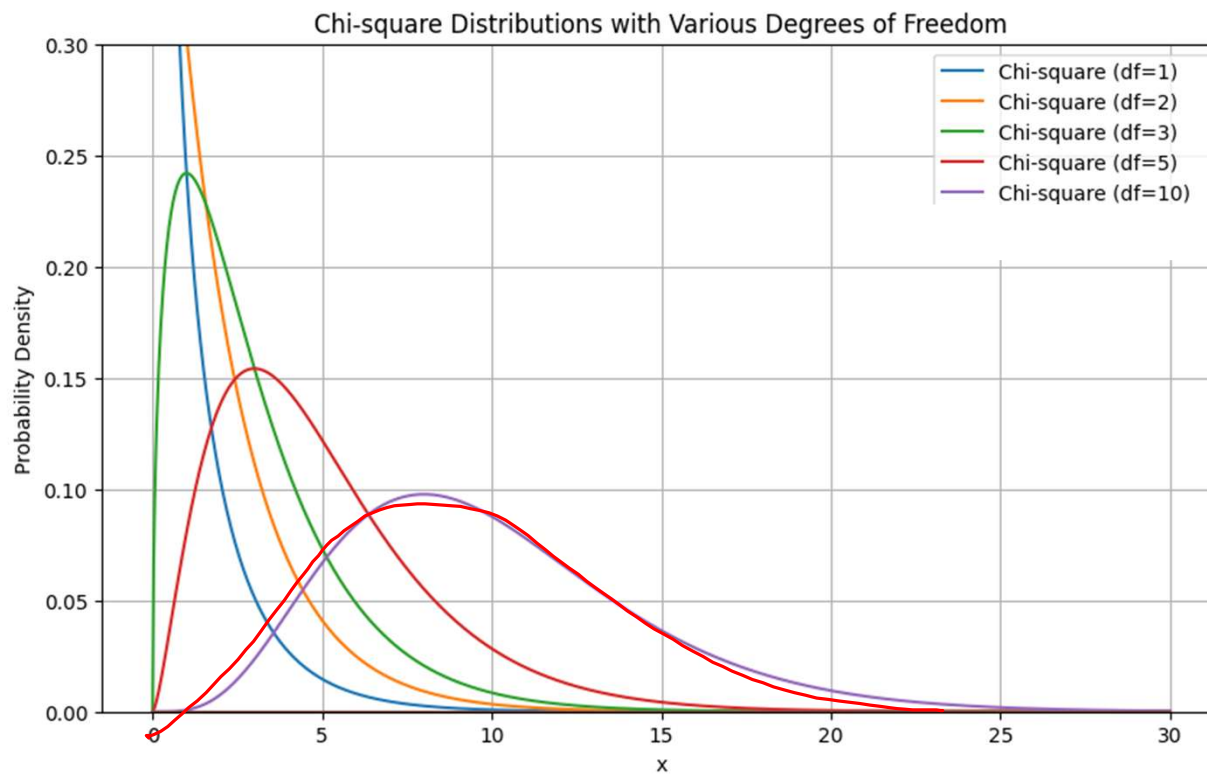
- ❖ The Chi-Square distribution is a continuous probability distribution.
- ❖ It is commonly used in hypothesis testing, especially in tests of independence and goodness of fit.
- ❖ The distribution is characterized by its degrees of freedom (df), which depend on the sample size.
- ❖ Chi-Square distribution only takes positive values because it's derived from squared values.

CHI-SQUARE DISTRIBUTION

Key Properties

- ❖ The shape of the Chi-Square distribution depends on the degrees of freedom (df).
- ❖ When df is small, the distribution is skewed to the right.
- ❖ As df increases, the distribution becomes more symmetric.
- ❖ The mean of the Chi-Square distribution is equal to the degrees of freedom (df).
- ❖ The variance is equal to $2 \times df$.
- ❖ Chi-Square values can be used to test the variability of the data relative to expected values.

CHI-SQUARE DISTRIBUTION



CHI-SQUARE DISTRIBUTION

- ❖ The probability density function (PDF) of the chi-square distribution with ν degrees of freedom is given by:

$$f(x; \nu) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\frac{\nu}{2}-1} e^{-x/2}, \quad x > 0$$

Where:

- x is the value of the chi-square variable.
- ν is the **degrees of freedom**.
- $\Gamma(\cdot)$ is the **Gamma Function** (which generalizes the factorial function for real and complex numbers).

CHI-SQUARE DISTRIBUTION

❖ The formula for the Chi-Square statistic is:

$$\chi^2 = \sum \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

❖ Where:

❖ O_i = Observed value

❖ E_i = Expected value

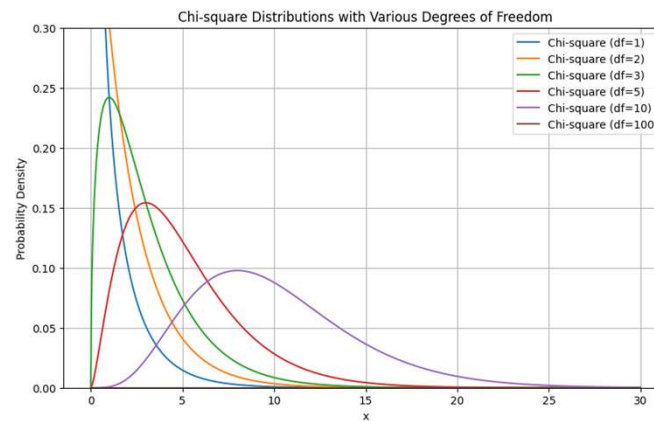
❖ Explanation:

❖ The Chi-Square statistic measures the discrepancy between the observed and expected values in categorical data.

❖ It is widely used in goodness-of-fit tests and tests of independence.

CHI-SQUARE DISTRIBUTION

- ❖ Key Points:
- ❖ The chi-square distribution is a special case of the gamma distribution.
- ❖ As v increases, the shape of the chi-square distribution becomes more symmetric, approaching a normal distribution.



F DISTRIBUTION

- ❖ The F distribution is a probability distribution used primarily in the analysis of variance (ANOVA) and hypothesis testing.
- ❖ The F-distribution is asymmetric and skews to the right.

Key Characteristics:

- ❖ Used for: Comparing variances between two populations.
- ❖ Degrees of Freedom: The F-distribution has two degrees of freedom (df_1 and df_2), one for each chi-square distribution.

F DISTRIBUTION

- ❖ The probability density function (PDF) of the F-distribution with degrees of freedom $df1$ and $df2$ is given by:

$$f(x; df1, df2) = \frac{\left(\frac{df1}{df2}\right)^{df1/2} x^{df1/2-1}}{\left(1 + \frac{df1}{df2}x\right)^{(df1+df2)/2} B\left(\frac{df1}{2}, \frac{df2}{2}\right)}$$

Where:

- $B(\cdot, \cdot)$ is the **Beta function**, which is related to the Gamma function.
- $df1$ and $df2$ are the degrees of freedom for the numerator and denominator, respectively.

F DISTRIBUTION

❖ F-distribution is the ratio of two independent chi-square distributions.

❖ Formula:

$$❖ F = \frac{\frac{S_1^2}{\frac{v_1}{S_2^2}}}{\frac{v_2}{v_2}} \sim F(v_1, v_2)$$

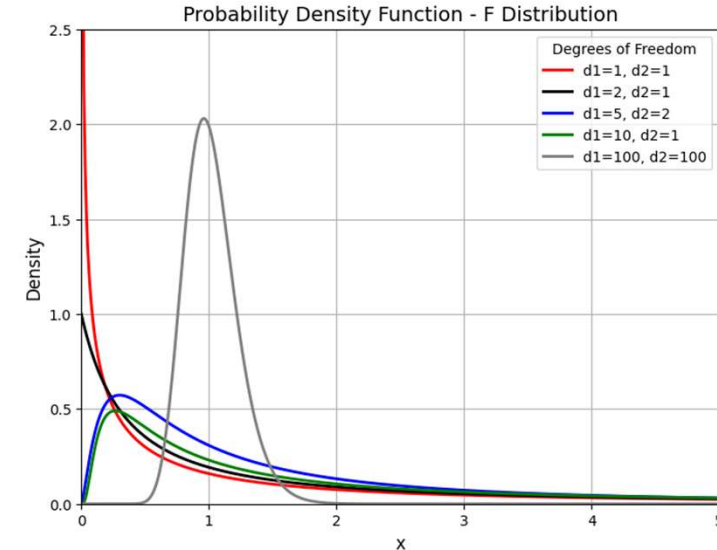
❖ Where:

- ❖ S_1^2 and S_2^2 are independent chi-square random variables.
- ❖ v_1 and v_2 represent the degrees of freedom.
- ❖ F follows the F-distribution with degrees of freedom v_1 and v_2 .

F DISTRIBUTION

Properties of the F Distribution

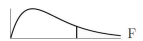
- ❖ Right-Skewed: The distribution is skewed to the right, with a long tail to the right.
- ❖ Non-Negative: Values of F-distribution are non-negative because variances (which are always positive) are compared.
- ❖ Mode: The mode shifts to the right as the degrees of freedom increase, making the distribution more symmetric.



F-DISTRIBUTION

- ❖ F-distribution tables exist for various significance levels (α), such as 0.01, 0.05, 0.10.
- ❖ These tables are used to find the critical F value for different degrees of freedom (df1 for the numerator, df2 for the denominator).
- ❖ To find the correct F-value, we need:
 - ❖ df1: Degrees of freedom for the numerator.
 - ❖ df2: Degrees of freedom for the denominator.
 - ❖ Alpha (α): The level of significance for the test (usually 0.05).

Table 6: F-table of Critical Values of $\alpha = 0.05$ for F(df1, df2)



	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.01	244.05	244.91
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66
4	7.71	6.94	6.59	6.39	6.25	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57

F-DISTRIBUTION

- ❖ Example: Find the critical F value for $df1 = 5$ and $df2 = 7$ at $\alpha = 0.05$.
- ❖ Check the F-distribution table for $\alpha = 0.05$.
- ❖ Locate the intersection of $df1 = 5$ (numerator) and $df2 = 7$ (denominator).
- ❖ The critical F value is $F(5,7, 0.05) = 3.97$.
- ❖ $df1$ and $df2$ are not interchangeable: The degrees of freedom for the numerator and denominator must be considered in the correct order.
- ❖ The critical F value is $F(7,5, 0.05) = 4.88$.

Table 6: F-table of Critical Values of $\alpha = 0.05$ for $F(df1, df2)$



	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56
6	5.95	5.08	4.66	4.42	4.28	4.18	4.11	4.04	3.99	3.96	3.90	3.84	3.78
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44
8	5.30	4.48	4.07	3.84	3.69	3.59	3.50	3.44	3.39	3.35	3.28	3.22	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57

RECIPROCAL RELATIONSHIP OF F-DISTRIBUTION

- ❖ The F-distribution has a reciprocal relationship when the degrees of freedom are swapped.

- ❖
$$F_{\alpha}(df1, df2) = \frac{1}{F_{1-\alpha}(df2, df1)}$$

- ❖ Example: Suppose you have $F_{(5,7,0.05)} = 3.95$.

- ❖ To find $F_{(7,5,0.95)}$, you can apply the reciprocal relationship:

- ❖
$$F_{(7,5,0.95)} = \frac{1}{F_{(5,7,0.05)}} = \frac{1}{3.95} \approx 0.25$$

- ❖ For most cases, you will be interested in the right tail area only.

